circuito a Apagando $F(t)$

$$F_{sa}(s) = \frac{1}{R(C_p + C_s)s + 1} F_o(s)$$

$$F_s(s) = \frac{C_s R_s + 1}{R(C_p + C_s)s + 1} F(s) + \frac{1}{R(C_p + C_s)s + 1} F_o(s)$$

$$F_s(s) = \frac{(C_s R_s + 1) F(s) + F_o(s)}{R(C_p + C_s)s + 1}$$

Circuito a - apagando $F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{sa}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$\alpha F(s) = R X(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$F(s) = \frac{X(s)}{\alpha} \; \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{\frac{X(s)}{\alpha} \; \frac{R(C_p + C_s)s + 1}{}} = \frac{\frac{X_s}{\alpha}}{\frac{X_s}{\alpha} \; \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = \frac{\alpha}{R(C_p + C_s)s + 1}$$

$$F_{s2}(s) = \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

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análisis del circuito por superposición

Circuito 1: Apagando $F_0(t)$

$x(t) = x_1(t) + x_2(t)$

$x(t) \rightarrow C_p \dfrac{dF_{s1}(t)}{dt}$

$x_1(t) = C_s \dfrac{d[F(t) - F_{s1}(t)]}{dt}$

$x_2(t) = \dfrac{F(t) - F_{s1}(t)}{R}$

$C_p \dfrac{dF_{s1}(t)}{dt} = C_s \dfrac{d[F(t) - F_{s1}(t)]}{dt} + \dfrac{F(t) - F_{s1}(t)}{R}$

$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \dfrac{F(s) - F_{s1}(s)}{R}$

$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \dfrac{F(s)}{R} - \dfrac{F_{s1}(s)}{R}$

$\left(C_p s + C_s s + \dfrac{1}{R}\right) F_{s1}(s) = \left(C_s s + \dfrac{1}{R}\right) F(s)$

$\dfrac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \dfrac{C_s R s + 1}{R} F(s)$

$F_{s1}(s) = \dfrac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} = \dfrac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$

Circuito 2: Apagando $F(t)$

$C_s s F_{s2}(s) + C_p s F_{s2}(s) + \dfrac{F_{s2}(s) - F_0(s)}{R}$

$\left(C_p s + C_s s + \dfrac{1}{R}\right) F_{s2}(s) = \dfrac{1}{R} F_0(s)$

$F_{s2}(s) = \dfrac{\frac{1}{R}}{C_p s + C_s s + \frac{1}{R}} + R = \dfrac{\frac{1}{R}}{R(C_p s + R C_s)s + 1} F_0(s)$